Sahaj Bhadja

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Summary

Forward Deployed Engineer with 2+ years turning ambiguous business needs into shipped **GenAI** applications, **workflow automation**, and integrations across **Python**, **React**, SQL, and Azure/AWS, presenting progress and value to stakeholders.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Web Technologies, Database Systems, Advanced NLP, Analysis of Algorithms

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Object-Oriented Programming, Data Retrieval, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Forward Deployed Engineer *Los Angeles, CA*

- Shipped a multi-tenant **AI education platform** on Azure SQL and Blob Storage with **row-level security** governing data access for 500+ users, cutting dashboard latency from 1.2s to 350ms.
- Built **data integrations** across FastAPI services, PostgreSQL, SQL Server, and cloud storage with validation and reconciliation, running 5K+ monthly jobs and 1K+ LLM jobs at 99%+ production reliability.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Shipped a **generative AI** grading application with React and Flask, combining GPT-4o, Claude, and Qwen-VL with OCR to automate evaluation for 150+ instructors and cut feedback turnaround 60%.
- Tracked per-call token usage and estimated **cost** across models with fallback routing for OCR reliability, keeping 10K+ daily requests cost-aware and stable while concurrent batches processed 1K+ files.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Automated cross-platform data-sharing workflows on microservices, cutting manual data entry 35%, while collaborating with Microsoft and GitHub **stakeholders** and running parallel builds on India and US servers in Agile sprints.
- Led a team modernizing **ETL** workflows on AWS Glue and S3, boosting pipeline efficiency 68%, cutting storage costs 15%, and reducing technical debt 20% in legacy integration code.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built a MERN and Azure SQL resource-management CRM with eight microservices for **resource allocation**, billing, leads, and notifications, synchronizing 2M+ records across locations with real-time WebSocket updates.
- Ran **observability** with Sentry, Azure Monitor, and Kibana, Snyk security scanning in Jenkins CI/CD, and analytics-driven changes that cut resource wastage 42% and raised lead conversions 15%.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Presented the end-to-end architecture to the client, translating **business requirements** into 12+ integration points, then delivered Flask booking, pricing, and payment APIs with sub-200ms latency across 50K+ records.
- Automated refund, dispute, and chargeback handling through idempotent payment webhooks, cutting manual work 70%, and built hourly Databricks **analytics** on service demand and revenue for client reporting.

Skills

Languages: Python, TypeScript, JavaScript, SQL, Java
Full Stack: React, Next.js, Node.js, FastAPI, Flask, REST APIs, Microservices
GenAI & Automation: Claude, GPT-4o, Prompt Engineering, RAG, OCR, LLM Orchestration, AI-Assisted Development, Workflow Automation
Data & Integration: PostgreSQL, SQL Server, Azure SQL, MongoDB, ETL Pipelines, AWS Glue, Databricks, Data Integration
Cloud & Observability: Microsoft Azure, AWS, Docker, CI/CD, Jenkins, Sentry, Azure Monitor, Kibana
Governance & Business: Row-Level Security, RBAC, Snyk, Requirements Gathering, Stakeholder Communication, Agile/Scrum

Projects

Manifest Intake and Compliance Automation | *Python, FastAPI, Claude, OCR, React, PostgreSQL, Azure*

- Designed a **GenAI** intake app that extracts fields from scanned waste manifests with OCR and Claude, validates them against **compliance rules**, routes exceptions to a human review queue, and logs auditable decisions.

HMIS – Hospital Management Information System Live | *TypeScript, MERN, ABDM APIs, Twilio, Sentry*

- Won Smart India Hackathon with a TypeScript workflow app using **consent-based** ABDM records access, appointment automation, Twilio alerts, and Sentry monitoring, cutting hospital administrative workload 30% across 100+ daily appointments.

Chat2SQL Code | *Java, TypeScript, SQL Server, PostgreSQL, Docker*

- Built an AI-assisted SQL and reporting workspace on Java and TypeScript supporting 10+ databases, generating schema-aware queries at 82% accuracy and AI-generated dashboards for faster **business intelligence**.